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RESEARCH REPORT

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Electrophysiological markers of poor versus superior math abilities in healthy individuals

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Abstract

Interindividual differences in the numerical ability of healthy adults have been previously demonstrated, mainly with tasks involving mental number line or size representation. However, electrophysiological correlates of superior versus poor arithmetic ability (in the healthy population) have been scarcely investigated. We correlated electric potentials with math performance in 13 skilled and 13 poor calculators selected from a sample of 41 graduate students on the basis of their poor or superior math abilities assessed through a timed test. EEG was recorded from 128 channels while participants solved 352 arithmetical operations (additions, subtractions, multiplications, divisions) and decided whether the provided solution was correct or incorrect. Overall skilled individuals correctly solved a higher number of operations than poor calculators and had faster response times. Consistently, the latency of frontocentral P300 component of event-related potentials (ERPs) peaked earlier in the skilled than poor group. The P300 was larger in amplitude to correct than incorrect solutions, but just in the skilled group, with a tendency found in poor calculators. Spearman's ρ correlation coefficient analyses showed that the larger P300 response was to correct arithmetic solutions, the better the performance; conversely, the larger the P300 amplitude was to incorrect solutions, the worse the performance. The results suggest that poor calculators had a less clear representation of arithmetic solutions and difficulty in quickly accessing it. This study provides a standard method for directly investigating math abilities throughout ERP recordings that could be useful for assessing acalculia/dyscalculia in the clinical population (children, elderly, brain-damaged patients).

KEYWORDS

arithmetic processing, Dyscalculia, event-related potentials, individual differences, number, P300, skills

Abbreviations: ADHD, Attention deficit and hyperactivity disorder; ANOVA, Analysis of variance; APA, American Psychological Association; EEG, Electroencephalogram; EOG, Electrooculogram; ITI, Intertrial interval; IQ, Intelligence quotient; ISI, Interstimulus interval; ERP, Event-related potential; fNIRS, Functional near-infrared spectroscopy; LP, Late positivity; μV, Microvolt; rIPS, Right intraparietal sulcus; RT, Reaction time; *SD*, Standard deviation; *SE*, Standard error.

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1 | INTRODUCTION

Neuroimaging studies have investigated how interindividual math abilities differentially affect the neural representation of arithmetic processing. Differences in the activation of the left angular gyrus (Grabner, Reishofer, Koschutnig, & Ebner, 2011; Menon et al., 2000), connectivity of the intraparietal sulcus (IPS; Park, Park, & Polk, 2012; Prado et al., 2011), activation of right dorsolateral prefrontal cortex (Ansari, Grabner, Koschutnig, Reishofer, & Ebner, 2011; Menon, Mackenzie, Rivera, & Reiss, 2002; Pesenti et al., 2001) and the left angular gyrus (Grabner et al., 2007) have been found across high and low math skill individuals. Consistently, rTMS stimulation of the left angular gyrus has been shown to disrupt numeric processing (Rusconi, Walsh, & Butterworth, 2005) that independent studies have shown to be involved in arithmetic processing (see Arsalidou and Taylor (2011) for a review). Again, Evans, Flowers, Napoliello, Olulade, and Eden (2014) observed hypo-activation of the left supramarginal gyrus in children with dyslexia during addition and subtraction operations. A role of the right angular gyrus/intraparietal sulcus (rIPS, supporting number representation, e.g., Proverbio, Bianco, & de Benedetto, 2018) in acalculia and arithmetic deficits has also been demonstrated in right-damaged patients (e.g., Benavides-Varela et al., 2017). Dyscalculia is very frequent in Turner syndrome (Bruandet, Molko, Cohen, & Dehaene, 2004; Mazzocco, 1998; Temple & Marriott, 1998). A deficit in mathematics, and particularly in arithmetic, has been documented in association with a bilateral parieto-occipital dysfunction (Reiss, Mazzocco, Greenlaw, Freund, & Ross, 1995).

While there are a few studies investigating ERP indices of arithmetic processing (e.g., Núñez-Peña, Cortiñas, & Escera, 2006; Núñez-Peña, Honrubia-Serrano, & Escera, 2005; Pauli, Lutzenberger, Birbaumer, Rickard, & Bourne, 1996; Pauli et al., 1994; Zhou et al., 2006, 2009), individual differences in mathematical skills reflected in event-related brain potentials have been seldom explored. The only exception is (as far as we know) Núñez-Peña, Gracia-Bafalluy, and Tubau's (2011) investigation, in which they compared brain potentials recorded in individuals with high and low math skill engaged in arithmetic tasks involving small (e.g., 3 + 4), medium (e.g., 7 + 8) and large numerical sets (e.g., 16 + 29). The authors found a late positivity (LP) that increased as task difficulty increased; but while LP did not change in amplitude when solving small versus medium problems in high skill individuals, it was greater to medium than to small problems in low-skill individuals, thus showing that the workload of medium problems was more challenging for them because of their reduced arithmetic skills.

Apart from these studies, mainly observing the effect of workload on numerical ability, the literature on neural markers or ERP indicants of math abilities in healthy adults is very scarce. In fact often subjects are tested not on arithmetic abilities per se, but on number related processes such as digit memory retrieval (Wang, Du, & Ma, 2017), magnitude representation/estimation abilities (e.g., deciding: whether visually presented digits (1–4, 6–9) are smaller or larger than 5 (Soltész, Szucs, Dékány, Márkus, & Csépe, 2007; also Dehaene, 1996); mental number line representation (e.g., indicating the magnitude or direction of numeric stimuli, from 11 to 99; Jastrzebski, Crewther, & Crewther, 2015). In a very recent study, the neural correlates of math abilities were explored through a combined fNIRS and ERP study (Artemenko, Soltanlou, Dresler, Ehlis, & Nuerk, 2018). Subjects of this study were recruited and assigned to high and low math ability groups based on an online screening of arithmetic abilities. The screening consisted of speeded additions, subtractions, multiplications and divisions with simple and difficult items, and the task for each operation was to solve as many items as possible within 2 min. The arithmetic task consisted of two-digit addition and subtraction problems with two operands. High performers showed increased frontal activation especially in the left inferior frontal gyrus associated with differences in slow waves at 1,000–1,500 ms at frontal sites.

To study individual differences in math abilities, we used the event-related potential (ERP) technique, providing useful indices of cognitive processes, including arithmetic calculations. Several studies have found an arithmetic N400 that behaves very similarly to the N400 evoked by words, as opposed to the P300 component, associated instead with correct and congruent math solutions. The N400 component is a large negative deflection peaking at about 400 ms over centro-parietal scalp areas. It has been related to the detection of an incongruence in semantic, lexical or conceptual processing, including arithmetic computing (Kutas & Iragui, 1998; Lau, Phillips, & Poeppel, 2008).

Niedeggen and Rösler (1999) investigated whether incongruent solutions of simple multiplication problems would elicit similar event-related brain potentials as inappropriate words in sentences and found larger N400s (as opposed to P300s) to incorrect than correct arithmetic solutions. Single-digit multiplication problems were used to construct correct and incorrect arithmetic equations for a verification task. The operands were presented one after another and completed with either the correct or an incorrect solution. A comparable event-related brain potential (ERP) effect was elicited by incongruous solutions in simple multiplication problems (Niedeggen, Rösler, & Jost, 2007). Correct solutions elicited a large P300 response, while incorrect solutions evoked a negativity between 300 and 500 ms with a maximum over the centro-parietal scalp, followed by a late positive component which was interpreted as a "positive syntactic shift" or P600 (Hagoort, Brown, & Groothusen, 1993), also found

in response to syntactic and musical violations (Osterhout, Holcomb, & Swinney, 1994; Patel, Gibson, Ratner, Besson, & Holcomb, 1998). A similar P600 was found by Núñez-Peña and Honrubia-Serrano (2004) in response to rule violations in arithmetic tasks.

The aim of the present study was to investigate the electrophysiological correlates of math abilities in the healthy population. Participants were required to quickly solve hundreds of operations (additions, subtractions, divisions, multiplications) of varying difficulty, in a verification task. The participants recruited for the EEG recordings were selected from a sample of 41 graduate students tested for their arithmetic performance: They were asked to solve or correct as many operations as possible in 2 min. The fastest and slowest individuals were attributed to the group of skilled and poor calculators, respectively.

On the basis of previous literature, we hypothesized that arithmetic incongruities (i.e., math operations paired to an incorrect result) would considerably reduce the amplitude of P300 responses, whose amplitude would be increased by correct solutions. P300 component (or P3b) is one of the most studied brain potentials linked to categorization and decisional processes (e.g., Johnson, 1988; Kirschner, Cruse, Chennu, Owen, & Hampshire, 2015; Polich, 2007) in addition to other functions, and it is always larger to congruent than incongruent pairs (e.g., Bach, Gunter, Knoblich, Prinz, & Friederici, 2009; Farwell, Richardson, Richardson, & Furedy, 2014; Kałamała, Szewczyk, Senderecka, & Wodniecka, 2018; Proverbio, Calbi, Manfredi, & Zani, 2014). It also reflects response certainty and subjective confidence in making a decision (e.g., Dröge, Fleischer, Schlesewsky, & Bornkessel-Schlesewsky, 2016). With these premises, we expected to find a poorer performance, both in terms of speed of response and percentage of correct responses in poor calculators, possibly associated with a later P300 component latency in this group. Indeed, individual differences for P300 latency are correlated with mental functionspeed, such that shorter latencies are related to superior cognitive skills (e.g., Amin, Malik, Kamel, Chooi, & Hussain, 2015; Emmerson, Dustman, Shearer, & Turner, 1989; Johnson, Pfefferbaum, & Kopell, 1985; Pelosi et al., 1992; Polich, 2007).

Furthermore, we hypothesized that, due to task difficulty, N400 response (reflecting expectancies' violation) would not be much pronounced, especially in poor calculators. On the other hand, since the task consisted in evaluating solution accuracy, we expected that P300 (supposedly indexing cognitive updating) might be larger than N400 in this study, especially in poor calculators who because of their inability to quickly access arithmetic facts might rely more on the verification of solutions provided, than forming specific expectancies about the solution. For this reason, the P300 component was identified and measured in the two groups as a possible marker of math abilities in this study.

2 | METHODS

2.1 | Participants

Volunteers were recruited among B.A. students graduated at the University of Milano-Bicocca. They earned academic credits for their participation. Forty-one students (28 males, 13 females, mean age = 24.49 ± 3.12) took part in the pretest selection. They were all right-handed as assessed through the administration of the Edinburg Inventory (Oldfield, 1971). Laterality scores were 85 for the skilled group and 83 for the poor group (scale = −100 to 100).

They studied in humanistic, statistical, economical or scientific fields (in the end, they were equally distributed across the two subgroups). Participants were administered a shortened version of *N* section of Kit of Factor-Referenced Cognitive Tests (Ekstrom, French, Harman, & Derman, 1976) relative to arithmetical abilities including the four fundamental operations: division, multiplication, subtraction and addition (in this order). The pretest was divided into four parts; for each of these four parts, candidates were required to resolve or correct as many operations as possible in 2 min. The pretest was administered in paper format in the presence of the investigator who controlled and timed the candidates. This task was chosen to gain empirical measure of the candidate's arithmetic abilities and to be able to select the (worst and best) students on the basis of their ability to quickly solve arithmetical operations. The mean number of operations correctly solved by the group within the allowed period of time was then computed. The mean score was 70.73. The individuals showing the best ($N = 13$) and worst performance ($N = 13$) were identified and subdivided into two groups (above and below the mean). The first subgroup was deemed "skilled calculators," the second "poor calculators." See Figure 1 for a distribution of scores across people belonging to the two subgroups. In order to avoid including subjects with potentially undiagnosed calculation disorders, it was preliminarily established to exclude participants with a performance below 2 *SD*s (29.76 or less), but no such case was found.

The EEG data of two males for each of the two groups were not included in the final grand average, for excessive ocular and muscular artifacts. The final sample had a mean age of 24.02 years (skilled group = 24.17 year [seven males, four females]; poor group 23.88 years [four males, seven females]).

It was assessed (through self-report) that subjects had never suffered brain damage, had no previous history of epilepsy and were under no current medications that could affect the electrical activity of the brain. None of the participants had been diagnosed as dyscalculic, dyslexic or had any learning or attentional disorder (e.g., ADHD; acalculia). The experiment was conducted with the approval of

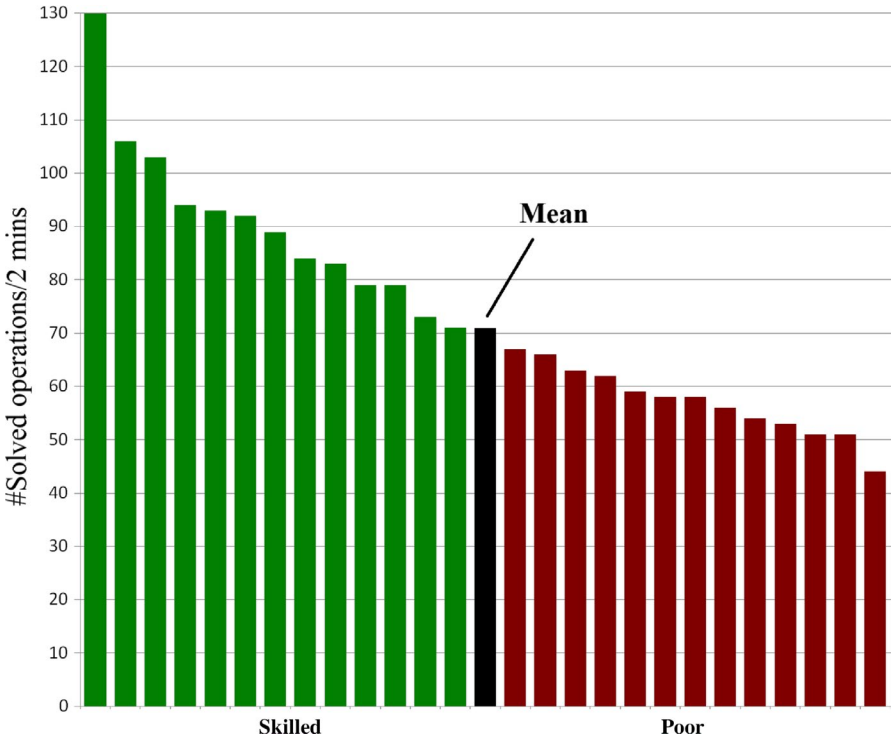

FIGURE 1 Average number of solved operations within 2 min by each individual subject. A shortened version of *N* section of Kit of Factor-Referenced Cognitive Tests (Ekstrom et al., 1976) was administered in paper and pencil format. The mean value was used to subdivide the population and consider the most skilled and the less skilled individuals. [Colour figure can be viewed at wileyonlinelibrary.com]

the Ethical Committee of the University of Milano-Bicocca (Prot. 0000273/14) and in compliance with APA ethical standards for the treatment of human volunteers (1992, American Psychological Association). Informed written consent was obtained from all subjects. All experiments were performed in accordance with relevant guidelines and regulations.

2.2 | Procedure

Participants were seated comfortably in a dark, acoustically and electrically shielded cubicle facing a screen located 114 cm from their eyes. They were instructed to gaze at the center of the screen, where a small red circle served as a fixation point, and to avoid any eye or body movements during the recording session. Participants rested their head against the back of their armchair and had to stay relaxed and perfectly still (while fixating the foveal point). No chin rest was used. Stimuli were presented in random order at the center of the screen in 10 experimental runs of 35–37 trials. Each run lasted about 165 s each. The experimental session was preceded by a training session that included two runs, one for each response hand, and containing stimuli discarded during the validation phase. The sequence presentation order varied across subjects. Each run featured a balanced number of correct and incorrect operations, presented in a randomized manner, and each session began with three warning stimuli ("READY," "SET" and "GO") lasting 700 ms. Each operation was presented for 1,500 ms and was followed by a random ISI of 100–150 ms. Results were flashed for 1,500 ms, and intertrial interval (ITI) was of 1,000 ms. Responses should be provided immediately after result presentation, within 1.5 s.

The task (a verification paradigm) required participants to decide as accurately and quickly as possible about the correctness or incorrectness of the operation's solutions by pressing a response key with their index finger (of the left or right hand), to signal that a solution appeared correct, and another key with their middle finger (of the left or right hand) to signal that it was incorrect. Previous literature has shown no significant difference in response speed between the two fingers (index vs. middle) in button press tasks (Lachnit & Pieper, 1990). The hand used was alternated during the recording session. Hand order and experimental runs were counterbalanced across subjects. At the beginning of each session, participants were reminded which response hand would be used.

2.3 | Stimuli and materials

Four hundred and forty arithmetic operations (divisions and subtractions) along with their results were selected as experimental stimuli. More (44 additions and multiplications) were added for a total of 484 arithmetic operations (242 correct and 242 incorrect) of various difficulty, but mentally solvable (at least by approximation) by anyone who had completed compulsory schooling.

Arithmetic operations were created according to the following criteria: They included integers between 0 and 2,000; contained a maximum of eight characters including the operator; each operation had to be followed by a whole solution

TABLE 1 Examples of operations followed by a correct (left column) or an incorrect (right column) arithmetic result

Correct		Incorrect	
Operation	**Result**	**Operation**	**Result**
4 + 5	9	1−5	5
9:1	9	0:35	4
38−29	9	58−48	4
63:7	9	120:12	3
69−60	9	40 + 0	4
90:10	9	48:19	2
485−476	9	50−40	5
459:51	9	294:30	2
558−549	9	201−195	5
990:110	9	1,400:500	3
19−11	8	10−7	4
16:2	8	9:6	3
36−28	8	16−10	3
48:6	8	46:30	2
50−42	8	84−80	3
24:3	8	767:476	1
83−75	8	60−51	4
728:121	8	570:280	1
500−492	8	75−66	4
2,000:250	8	910:910	2

between 0 and 10. For this reason, an asymmetrical number of arithmetic operations was used (mostly divisions and subtractions), and arithmetic operation type could not be therefore a factor of observed variability.

- Half of the operations had to be followed by a correct solution, the other half by an incorrect solution. This was true for each of the four operations.
- Incorrect solutions had to be diversified and clearly different from the correct result to allow approximate calculations, given the fast rate of stimulation and the temporal length of the test requiring mental concentration for a prolonged period of time (half an hour of EEG recording, including pauses). Indeed, it has been shown that the more two numerals are farther apart (e.g., 2 vs. 8) the easier is to compare them, as opposed to when they have a similar numerical value (e.g., 2 vs. 3) than closer ones. This effect is known as distance effect (Dehaene, Dupoux, & Mehler, 1990; Henik & Tzelgov, 1982; Tzelgov, Meyer, & Henik, 1992). For this reason, the distances of the incorrect solutions were matched across operation types.

By observing no operation in particular, we wished to annul the effect of a specific operation, thus gaining information about arithmetic abilities per se. Furthermore, since the experimental purpose was not to discriminate between different operations, numerical codes were not associated differently with stimuli linked to different operations, so that EEG trials relative to different operations were average together.

Stimuli were typed in Times New Roman font and yellow (FFFF00) color on a gray (424246) background (to avoid after effects) and were presented at the center of the screen. Overall stimulus size depended on the number of ciphers but did not exceed 2.5° of visual angle.

2.4 | Stimuli validation

For stimulus validation, a sample of 12 judges (five women and seven men) was recruited. Judges had a mean age of 34.08 years ($SD = 11.5$) and had 15 years of schooling on average ($SD = 3.33$). In order to balance all the operations per difficulty (across the correct and incorrect conditions), judges were presented with all the stimuli, organized in pseudorandom sequences, with each operation followed by a correct (or incorrect solution), and were asked to evaluate the stimuli for subjective difficulty. They were required to first solve and afterward rate each problem by using a three-point Likert scale. In particular, they had to answer:

- 0 for all solutions that appeared to be clearly correct or clearly incorrect;
- 1 for all solutions that they had recognized with difficulty to be correct or incorrect, or that had left them in doubt about the answer to be given;
- 2 for all operations that appeared to be unsolvable or too confusing.

On the basis of this procedure, operations that were designated by at least ¼ of the judges as too difficult to be mentally solved (score 2) were eliminated. Furthermore, operations that were considered as clearly solvable by 100% were eliminated as too simple (score 0), except for additions and multiplications (only 10% of all operations). The final group of operations consisted of 352 stimuli, 176 followed by correct results and 176 by an incorrect result. An ANOVA proved that the degree of difficulty was perfectly balanced across the two classes: correct versus incorrect ($p = 0.97$). Example of stimuli is provided in Table 1.

2.5 | EEG recording and data analyses

The EEG was continuously recorded using the software package EEvoke (ANT, Enschede) from 128 scalp sites arranged according to the international 10–5 system (Oostenveld & Praamstra, 2001) at a sampling rate of 512 Hz. Horizontal and vertical eye movements were also recorded. Average mastoids served as the reference lead. The EEG and

electrooculogram (EOG) were amplified with a half-amplitude bandpass of 0.016–100 Hz. Electrode impedance was maintained below 5 kΩ. EEG epochs were synchronized with the onset of the operation result. A computerized artifact rejection criterion was applied before averaging to discard epochs in which eye movements, blinks, excessive muscle potential or amplifier blocking occurred. The artifact rejection criterion was a peak-to-peak amplitude exceeding 30 μV, and the rejection rate was ~5%. ERPs were averaged off-line from −100 to 1,500 ms after stimulus onset and analyzed via software EEProbe (ANT, Enschede).

Event-related potential components were identified and measured with reference to the average baseline voltage over the interval of −100 to 0 ms at the sites and latencies at which they reached their maximum amplitudes. EEG epochs associated with an incorrect or rejected behavioral response were excluded. The mean area amplitude of P300 component was measured at Fz, FCz and Cz sites within the 280–430 ms time window.

Event-related potential data were subjected to a 3-way multifactorial repeated-measures ANOVA with 1 between-subject factor (Group: Skilled vs. Poor) and 2 within-subjects factors: correctness (correct, incorrect) and electrode (3 levels: Fz, FCz and Cz). Multiple comparisons of means were performed with Tukey's post hoc test.

Spearman ρ correlations were performed between individual P300 latency values and response times, individual P300 amplitude values and response times, and individual P300 amplitude values and percentage of correct responses (Hits), regardless of group and electrode sites. These correlations were planned on the basis of the known relationship between P300 latency and response speed, as well as that of P300 amplitude and accuracy in performance (e.g., Duncan-Johnson & Donchin, 1980; Houlihan, Campbell, & Stelmack, 1994). The correlation between P300 amplitude and response speed (not theoretically justified) was performed to test the reliability of the statistical method, and a lack of significance was indeed predicted for this analysis.

Reaction times (RTs) that exceeded the mean value ±2 *SD* were discarded, which resulted in a rejection rate of 2%. Hit rate percentages were converted to arcsine values and used for analysis of variance. Because percentage values do

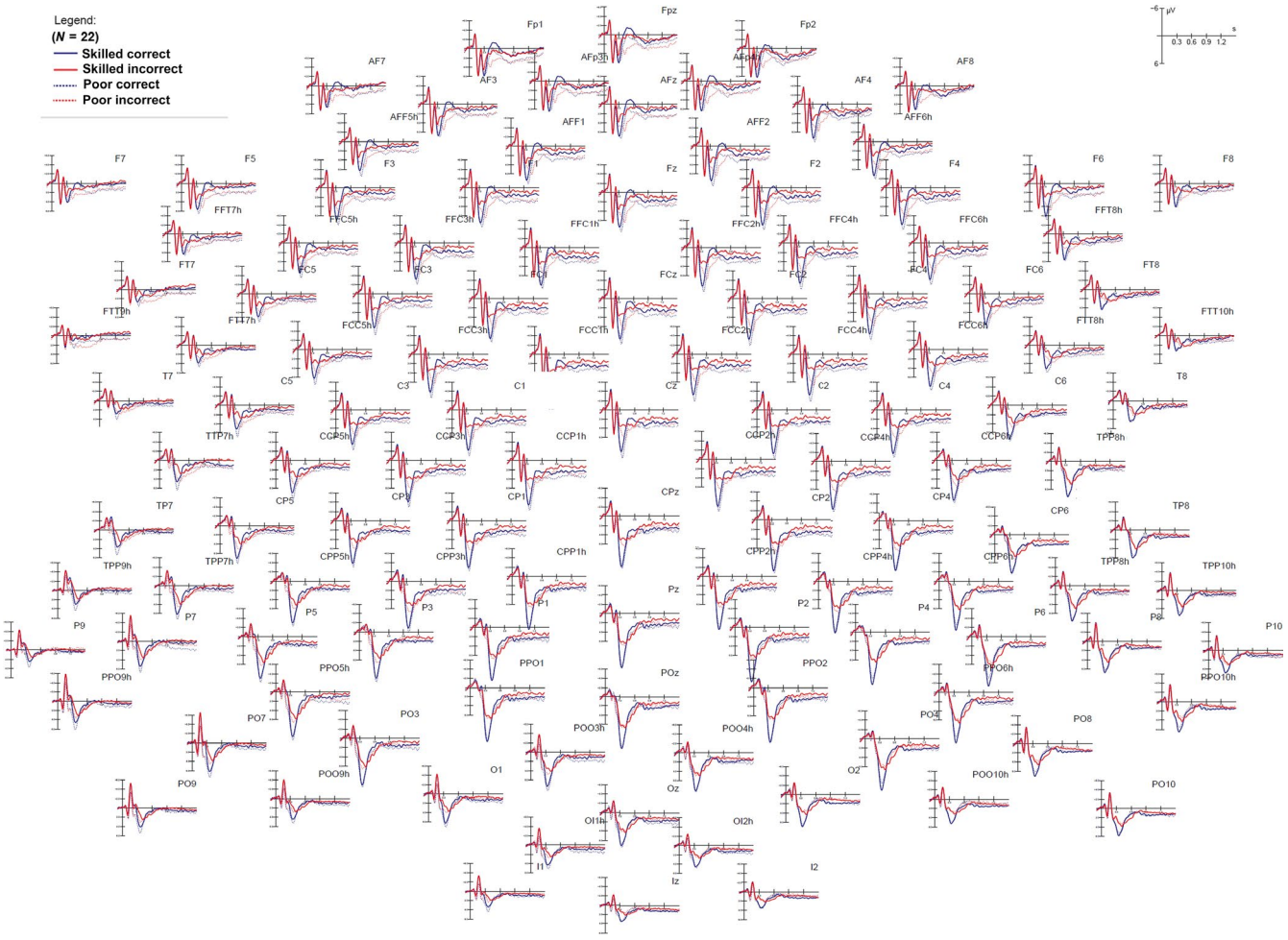

FIGURE 2 Grand-average event-related potentials recorded from 128 scalp sites in the two groups of skilled and poor calculators as a function of result correctness. It can be noted a large group difference especially affecting the latency of fronto-central P300 and its amplitude in response to incorrect results. [Colour figure can be viewed at wileyonlinelibrary.com]

not exhibit homoscedasticity (e.g., Snedecor & Cochran, 1989), which is necessary for ANOVA, the data were converted into arcsine values. In fact, the distribution of percentages is binomial, whereas the arcsine transformation of the data makes the distribution normal. Both RTs and hit percentages were subjected to separate multifactorial repeated-measures ANOVA analyses with 1 between-subject factor (group: skilled vs. poor) and 2 within-subjects factors: correctness (correct, incorrect); response hand (left, right).

The Greenhouse–Geisser correction was applied to compensate for possible violations of the sphericity assumption associated with factors which had more than two levels. The epsilon (ε) values and the corrected probability levels (in the case of epsilon <1) are reported. Partial eta-squared values were systematically provided to estimate effect sizes.

3 | RESULTS

Figure 2 shows grand-average ERPs recorded from 128 sites in the two groups of participants and as a function of

TABLE 2 Mean amplitude values of P300 (along with *SE*) recorded in the two groups as a function of trial correctness and electrode site

GROUP	Correctness	Electrode	Mean in µV	*SE*	Statistical significance, *p*
POOR	CORRECT	Fz	5.57	0.81	0.179
POOR	INCORRECT	Fz	4.05	0.66	
POOR	CORRECT	FCz	6.07	0.98	0.174
POOR	INCORRECT	FCz	4.48	0.77	
POOR	CORRECT	Cz	6.28	1.07	0.073
POOR	INCORRECT	Cz	4.44	0.89	
SKILLED	CORRECT	Fz	5.47	0.67	0.018
SKILLED	INCORRECT	Fz	3.53	0.55	
SKILLED	CORRECT	FCz	6.58	0.81	0.001
SKILLED	INCORRECT	FCz	3.74	0.64	
SKILLED	CORRECT	Cz	7.09	0.89	0.0002
SKILLED	INCORRECT	Cz	3.56	0.74	

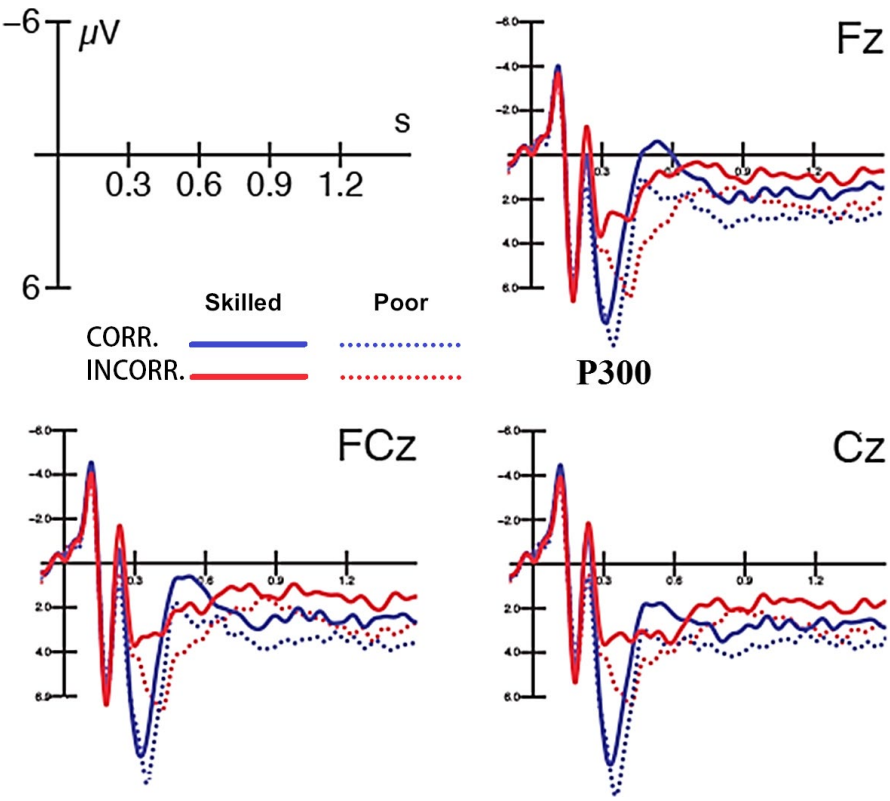

FIGURE 3 Grand-average event-related potentials recorded from midline frontal (Fz), fronto-central (FCz) and central (Cz) sites in response to correct and incorrect results, in the two groups. [Colour figure can be viewed at wileyonlinelibrary.com]

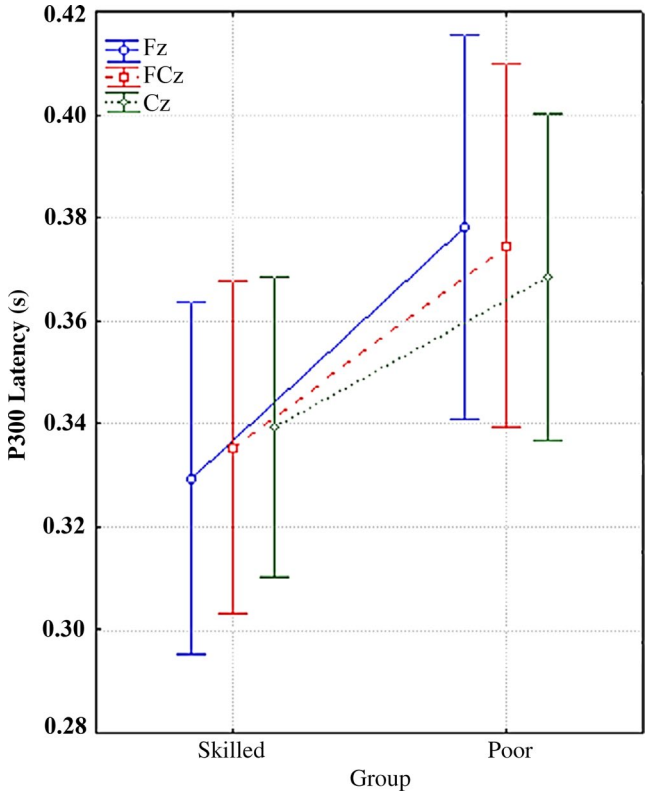

FIGURE 4 Mean values of P300 latency recorded at midline frontal (Fz), fronto-central (FCz) and central (Cz) site in the skilled and poor group of calculators. P300 peaked consistently earlier in the skilled group, regardless of other factors. [Colour figure can be viewed at wileyonlinelibrary.com]

result correctness. A prominent fronto-central P300 component, whose amplitude and latency varied as a function of group, result correctness, and in interaction with these factors, was noted. N400 was not visible in poor calculators for the reason previously explained (task difficulty and lack of expectations).

3.1 | Behavioural results

The analysis of correct responses showed the significance of group factor [$F_{1,20} = 9.55$; $p < 0.006$; $\varepsilon = 1$; $\eta_p^2 = 0.32$] with a better performance in the skilled (71%, $SE = 1.32$) than the poor group (65.18%, $SE = 1.32$). The correctness factor also yielded significance, with a higher rate of correct categorization of incorrect (71.6%, $SE = 1.65$) than correct solutions (64.53%, $SE = 1.61$) [$F_{1,20} = 27.14$; $p < 0.001$; $\varepsilon = 1$; $\eta_p^2 = 0.58$]. The analyses of RTs showed a significant group effect with skilled calculators responding much faster (561.48 ms, $SE = 34.6$) the poor calculators (729 ms, $SE = 34.6$) [$F_{1,20} = 4.4$; $p = 0.04$; $\varepsilon = 1$; $\eta_p^2 = 0.18$]. Also, significant was the correctness factor [$F_{1,20} = 43.8$; $p < 0.001$; $\varepsilon = 1$; $\eta_p^2 = 0.69$], with faster RTs to correct (613 ms, $SE = 25$) than incorrect solutions (695 ms, $SE = 26$). The hand factor also showed significance [$F_{1,20} = 6.31$; $p < 0.02$; $\varepsilon = 1$; $\eta_p^2 = 0.24$], with faster RTs emitted with the right (639 ms, $SE = 33$) than the left hand (669 ms, $SE = 38$).

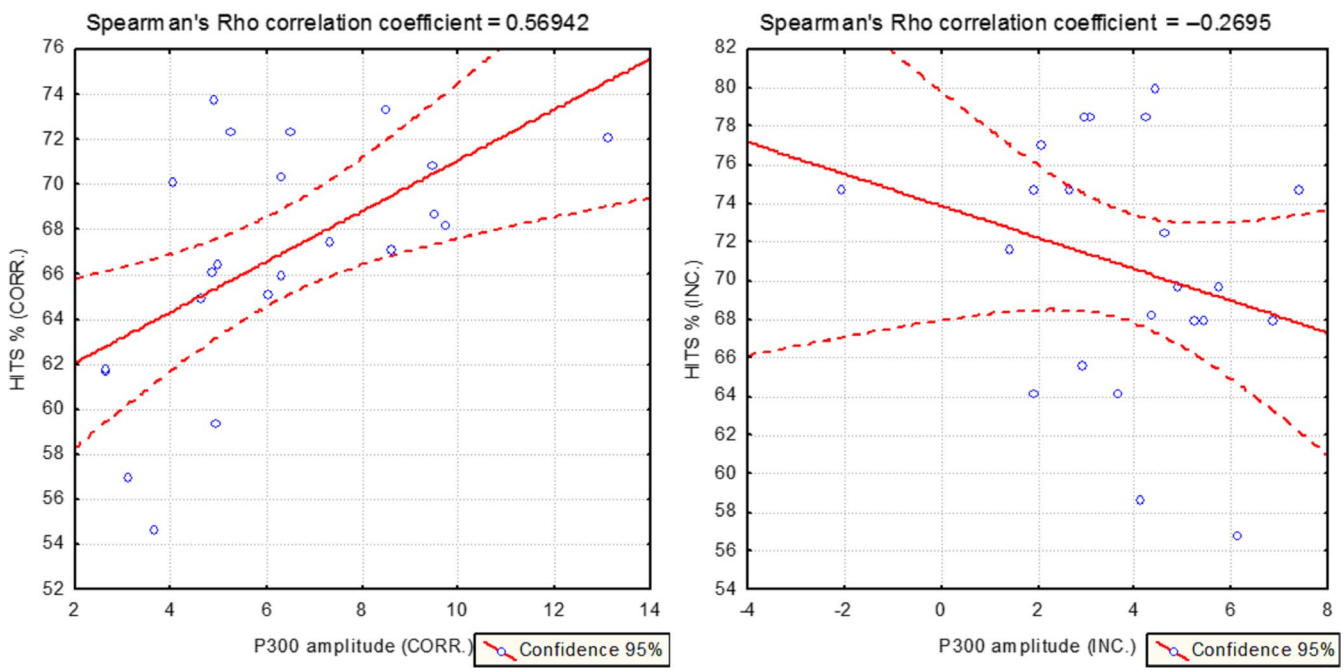

FIGURE 5 (Left) Spearman's ρ correlation between individual P300 amplitude values elicited by correct solutions and hit percentages (for correct solutions). The values were averaged across Fz, FCz and Cz electrode sites. (Right) Spearman's ρ correlation between individual P300 amplitude values elicited to incorrect solutions and hit percentages (for incorrect solutions). Again, the values were averaged across Fz, FCz and Cz electrodes. A larger P300 to incorrect solutions was associated with a worse math performance. [Colour figure can be viewed at wileyonlinelibrary.com]

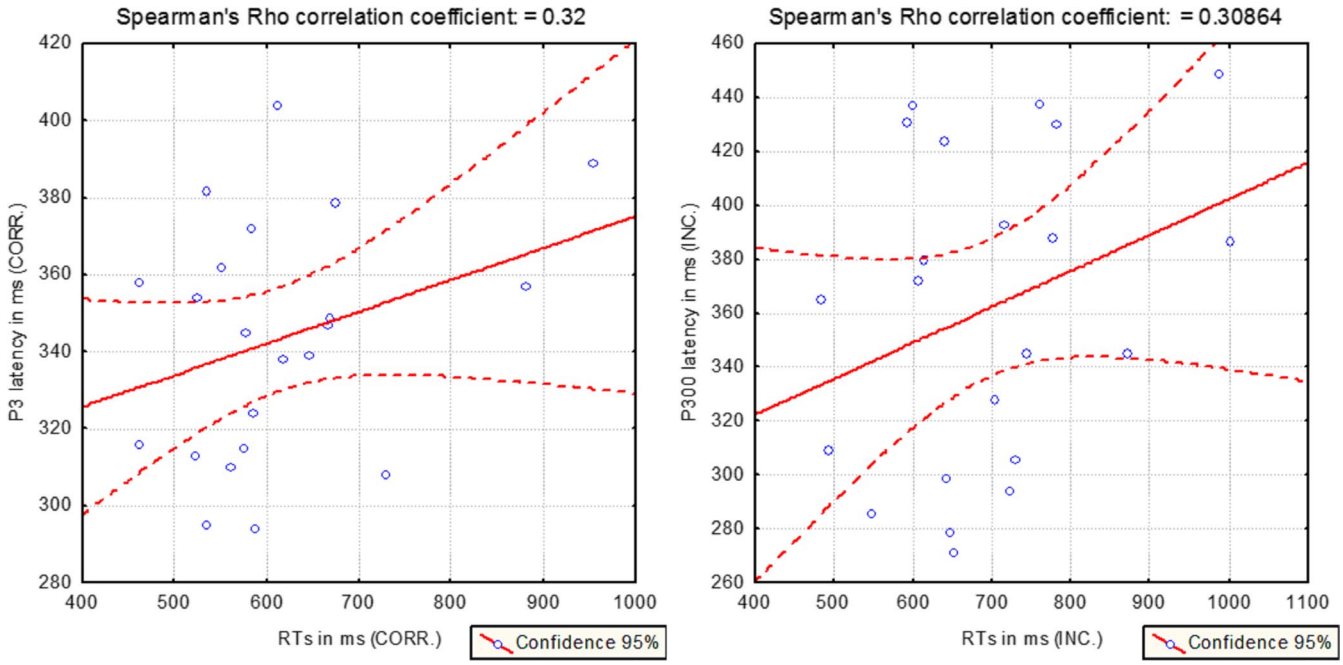

FIGURE 6 (Left) Spearman's ρ correlation between individual P300 latency values recorded for correct solutions and response speed relative to correct solutions. The values were averaged across Fz, FCz and Cz electrodes. (Right) Spearman's ρ correlation between individual P300 latency values recorded for incorrect solutions and response speed relative to incorrect solutions. Overall, the earlier P300 response peaked, and the earlier, the manual response was emitted. [Colour figure can be viewed at wileyonlinelibrary.com]

3.2 | Electrophysiological results

The ANOVA performed on P300 mean area amplitude showed the significance of correctness [$F_{1,20} = 23.7$; $p < 0.001$; $\varepsilon = 1$; $\eta_p^2 = 0.54$] with a much larger P300 to correct (6.17 μV, $SE = 0.59$) than incorrect solutions (3.96 μV, $SE = 0.47$). The further significance of correctness × electrode [$F_{2,40} = 12.0$, $p < 0.001$; $\varepsilon = 0.61$, adjusted p value = 0.001; $\eta_p^2 = 0.38$] indicated larger P300s to correct solutions over central (Cz: 6.68 μV, $SE = 0.69$) as opposed to frontal sites (Fz: 5.51 μV, $SE = 0.52$).

The interaction of group × correctness × electrode ($F_{2,40} = 5.40$, $p < 0.008$; $\varepsilon = 0.61$, adjusted p value = 0.02; $\eta_p^2 = 0.21$] was also found to be significant. Post hoc comparisons showed no group difference in P300 response to correct stimuli (p ranging from 0.94 to 0.77). However, P300 was much smaller to incorrect versus correct solutions in the skilled group, with no correct versus incorrect significant difference in the poor calculator groups, but a tendency of $p = 0.07$ just at Cz site (see values and post hoc comparisons provided in Table 2). This effect can be very well appreciated in Figure 3.

The ANOVA performed on P300 latency values showed the significance of group [$F_{1,20} = 6.26$, $p < 0.02$; $\varepsilon = 1$; $\eta_p^2 = 0.15$] indicating earlier P300s in skilled calculators (334 ms, $SE = 10$) than poor calculators (373 ms, $SE = 11$), as shown in Figure 4. The interaction of group × electrode [($F_{2,40} = 4.6$, $p < 0.016$; $\varepsilon = 0.83$, adjusted p value = 0.02; $\eta_p^2 = 0.19$] indicated faster P300 latency for the skilled calculator group especially at Fz electrode site.

Overall, there was no significant correlation between individual values of P300 amplitude and individual mean response times (Spearman's ρ correlation coefficient = −0.16). Similarly, no significant correlation was found between P300 amplitude to incorrect solutions and RTs to incorrect solutions (Spearman's ρ correlation coefficient = 0.06). On the other hand, a strong positive correlation ($p < 0.01$) existed between individual P300 amplitude values and hit percentages to correct solutions (Spearman's ρ correlation coefficient = 0.57). Conversely, a significant inverse correlation ($p < 0.05$) existed between P300 amplitude to incorrect solutions and hit percentages to incorrect solutions (Spearman's ρ correlation coefficient = −0.27): the larger the P300 to incorrect solutions, the lower the Hit percentage. In other words, the error rate due to miscategorizations (i.e., considering correct a mistake) increased as P300 to incorrect solutions increased (see Figure 5).

Further correlations were performed between P300 latency and response times. A significant positive correlation existed ($p < 0.05$) between the latency of P300 to correct solutions and RTs to correct solutions (Spearman's ρ correlation coefficient = 0.32). Again, a positive correlation existed ($p < 0.05$) between the latency of P300 to incorrect solutions and RTs to incorrect solutions (Spearman's ρ correlation coefficient = 0.31) as shown in Figure 6.

4 | DISCUSSION

The aim of this study was to investigate whether ERP potentials, and in particular P300 response linked to decisional processes, might reflect interindividual differences in the math abilities of a representative sample of healthy graduate students.

Overall, response times were faster when the right hand was used, which suggests a prevalent left-hemispheric involvement in arithmetic calculations (Altenmüller, Jung, Winker, & Landwehrmeyer, 1989), although the crucial role of the right hemisphere in number processing has also been demonstrated. For example, Proverbio, Bianco et al. (2018) found no hand asymmetry when subjects compared Arabic stimuli (for equality) but a right hand (left hemispheric) advantage when they compared numbers in verbal format. In this study, both skilled and poor calculators showed a right-hand advantage with no group difference in lateralization.

Behavioral results showed that individuals with high math abilities provided faster responses compared to individuals with low math abilities, similar to that found by Artemenko et al. (2018). This was associated with better accuracy (higher hit percentages) and to an overall earlier fronto-central P300 latency in the group of skilled calculators. Interestingly, having high math skills was also associated with smaller P300s to incorrect solutions, and, vice versa. Having poor math skills led to larger P300s to incorrect solutions, as demonstrated by significant Spearman correlations. Since in match/mismatch paradigms P3/N400 potentials reflect correct/congruent versus incorrect/incongruent categorizations (e.g., Farwell et al., 2014; Kałamała et al., 2018; Proverbio & Zani, 2003), it might be argued that low math skilled individuals had reduced abilities to reject incorrect solutions. This hypothesis is supported by Spearman correlations showing an inverse relationship between performance and P300 amplitude to incorrect solutions. The finding of a larger P300 to correct than incorrect arithmetic solutions is in accordance with previous electrophysiological literature (El Yagoubi, Lemaire, & Besson, 2003; Niedeggen & Rösler, 1999; Núñez-Peña & Escera, 2007; Núñez-Peña & Suárez-Pellicioni, 2012; Szűcs & Csépe, 2005). The reduced P300 amplitude in response to incorrect solutions corroborates many findings reporting enhanced negativities to incorrect solutions (Szűcs & Csépe, 2005; Wang, Kong, Tang, Zhuang, & Li, 2000). Niedeggen and Rösler (1999) found that incorrect solutions of multiplications elicited a negativity (named arithmetic N400) starting at about 300 ms poststimulus, which was followed by a centro-parietal late positivity. They interpreted these biphasic responses as indicating a conflict between the operation presented and the solution offered (see also Núñez-Peña et al., 2011).

Our hypothesis is that the low math ability individuals mainly relied on a strategy based on a plausibility check of the results, while the most capable individuals more often relied on certain calculations or on the re-evocation of arithmetic facts (for example, results of simple multiplication problems (e.g., 6×8) are retrieved from declarative memory as well-established facts, i.e., Niedeggen and Rösler (1999). The second strategy would have led to a clearer representation of the numerical result, and to a more distinguishable bioelectrical effect between correct and incorrect trials, which was actually found for skilled calculators. At this regard, it has been found that recruitment of the left angular gyrus during arithmetic problem solving underlies individual differences in math abilities: Its activation would index a stronger reliance on automatic, language-mediated processes in more competent individuals. Grabner et al. (2007) investigated through fMRI whether brain activation was modulated by individual differences in math abilities during mental calculation. Two groups of students were selected from a larger pool based on their performance on arithmetic (quite similar to the present study) and divided into groups of skilled versus poor calculators. The task required them to verify the correctness of provided multiplication solutions. The results showed that individuals with higher mathematical competence displayed stronger activation of the left angular gyrus (lAG), which has a crucial role in the retrieval of arithmetic facts (Grabner et al., 2009). Therefore, the hypothesis can be advanced that P300 response elicited to correct solutions in the present study might reflect the activity of lAG during retrieval of arithmetic facts, differently in skilled versus poor calculators. Indeed, several neuroimaging studies have provided evidence of a contribution of AG in generating P300 responses to targets. A correlation between white matter underneath the angular gyrus and P300 amplitude to targets has been demonstrated by Fusar-Poli et al. (2011), while significant increases in fMRI signal for both auditory and visual targets were observed in the supramarginal gyrus by Linden et al. (1999). Again Sabeti, Katebi, Rastgar, and Azimifar (2016) found that the angular gyrus (BA39) was one of the most prominent sources of P300 in memory processes.

In an interesting study, Luo, Liu, He, Tao, and Luo (2009) explored the electrophysiological markers of shortcut calculation strategy (i.e., rounding) and non-shortcut calculation strategy (i.e., exact calculation) in performing addition using mental arithmetic. It can be assumed that approximate versus exact calculations are associated with a smaller and greater degree of response/output certainty, that were linked to larger P220 and N400 responses in that study. Other evidence suggests that mental calculation is based on a dual process involving, on one hand, a visual mental imagery resolution strategy, and, on the other hand, a number-specific working memory (Zago et al., 2001), but their involvement might depend on the specific math operation to be solved (e.g., $20 + 20$ vs.

6 × 8). In the end, further research is needed to understand their relevance in explaining individual math skills.

In our study, the overall reduced negativity to incorrect solutions found in poor calculators suggests that, because of the task's difficulty, low math ability individuals mainly relied on a strategy based on a plausibility check of the solutions, rather than forming an expectation. This result is different from that found in other studies using much simpler arithmetic operations based, for example, on the well-known multiplication table (e.g., Niedeggen & Rösler, 1999), and in which incorrect solutions elicited N400 responses.

In conclusion, the present findings indicate that the latency and amplitude of P300 component evoked by the solution of a mathematical operation can be considered as reliable electrophysiological markers of superior versus poor math abilities in healthy individuals. Furthermore, data analysis suggests that the P300 component might be efficaciously used to assess mathematical comprehension and understanding in individuals unable to provide a written or vocal response (e.g., locked-in or brain-damaged patients). Furthermore, it can be universally used regardless of the native language spoken, or of language proficiency because it does not require the access to number phonological representation, being a YES/NO verification paradigm.

One limit of this study is that no other measure (such as brain connectivity, I.Q., verbal fluency, introversion–extroversion trait, etc.) was performed or correlated with the performance of individual subjects. Therefore, we ignore whether skilled calculators were more introverted (as shown for example by Proverbio, De Benedetto, Ferrari, & Ferrarini, 2018) or had thicker white matter. For example, children with superior math abilities have been shown to possess a higher degree of intrinsic functional connectivity of the hippocampus with dorsolateral and ventrolateral prefrontal cortices and the basal ganglia (Supekar et al., 2013). In our study, poor and skilled calculators had the same age and cultural status and had a degree in humanistic, scientific or technical fields (without differences between groups). The only mild difference concerned female individuals who were overrepresented in the poor math skills group. However, the study aim was to identify electrophysiological markers of math skills in healthy individuals regardless of other variables, such as the contributing factors to explain low mathskills.

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CONFLICT OF INTEREST

The authors declare no competing financial interests.

DATA ACCESSIBILITY

Anonymized data and details about preprocessing/analyses will be made available to colleagues if requested.

AUTHOR CONTRIBUTIONS

AMP conceived and planned the experiment. MC prepared the stimuli and carried out the EEG recordings. MC performed statistical analyses and contributed to the interpretation of the results. AMP interpreted the data and took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis and manuscript.

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